



Chapter 2 — Formation

How a predictor became an assistant

A model that only scores text continues your question instead of answering it

animation

the running question typed into a model that has had nothing added — it continues with more questions

1600 x 422 px

The question, throughout this chapter: *What does lowering the water/cement ratio do to compressive strength?*

observed

Substrate

Formation

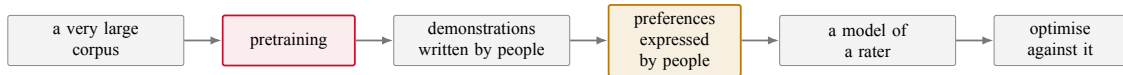
Control

Measure

Failure

Limits

Four stages turn a predictor into something that answers

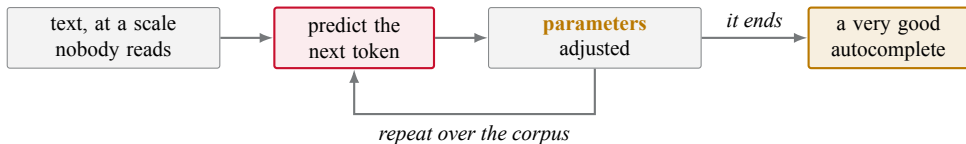


Every stage after the first is a person's judgement, written down and turned into a number. **Each name on this diagram is defined on the frame that expands it**, and the chapter walks the diagram once, left to right, stopping at each stage to ask what the job actually looks like.

derived



Pretraining runs one objective over a very large corpus, and then it stops



No labels

the text supervises itself, which
is why the corpus could be so
large

Knowledge cutoff

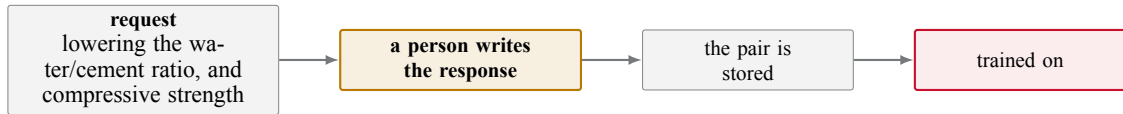
the date it stopped, and a
consequence rather than a policy

The corpus decides

where it is strong, and where it
is not

derived

People write the answer that should have come out, and the model is trained on the pair



Supervised fine-tuning

the model learns the *shape* of an answer — that a question gets answered

System prompt

standing text placed before your message, which this stage trains it to treat as instruction

Cheap in data terms

and it does most of the visible work of turning autocomplete into an assistant

[V] Ouyang, L., Wu, J., Jiang, X., et al., 2022. *NeurIPS 2022*, arXiv:2203.02155.

The person repairs the text as well as judging it, against a written specification

SCHEMATIC — the specification and both drafts are invented for this course.

“What does lowering the water/cement ratio do to compressive strength?”

draft

Lowering the water/cement ratio generally increases compressive strength — because less water means less excess to evaporate — so you'll typically see fewer voids forming, though it can vary a bit by mix design.

hedges removed

generally / typically / a bit

second person removed

“you'll”

one sentence per idea

the dash aside becomes two sentences

after the edit

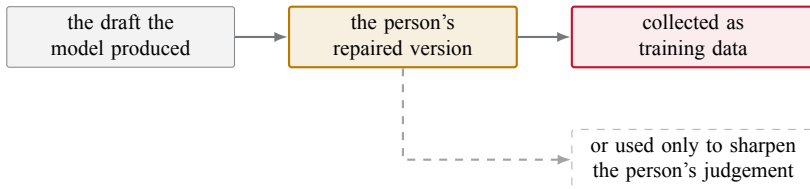
Lowering the water/cement ratio increases compressive strength.
Less water leaves less excess to evaporate.
Fewer voids form as that excess leaves.
Fewer voids raise the strength of the hardened concrete.

The rater does not only judge. They repair, to a written specification.

The person doing this is a **rater** — paid, working to a written instrument, usually with no contact with the people who wrote it. The specification is not about being right. It is about punctuation, hedging, person, sentence length and the shape of a list.

instrument (Three of five documents.)

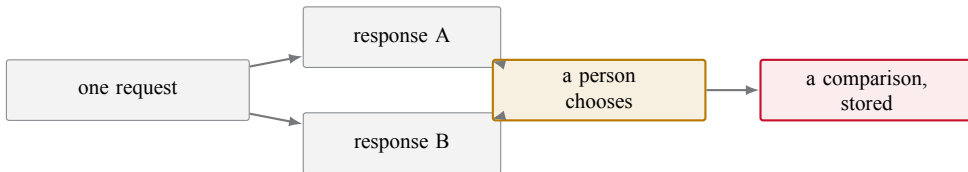
In two of the five instruments, the repaired text is what gets collected



The labour recurs in three documents. The repaired text entering the data is a separate claim on weaker evidence, and in one instrument the repaired version is explicitly not trained on.

instrument (Two of five documents for collection; three of five for the labour.)

Demonstrations cannot cover everything, so preferences are collected instead



Writing the ideal answer to every possible request is impossible. Judging which of two answers is better is not.

[V] Stiennon, N., Ouyang, L., Wu, J., et al., 2020. *preprint — venue unconfirmed*, arXiv:2009.01325.

A rating task is a rubric whose dimensions conflict

SCHEMATIC — instrument invented for this course.

rating one response to "What does lowering the water/cement ratio do to compressive strength?"

dimension	criterion — judged from the response alone, never the prompt	1-5				
truthfulness	every claim about material behaviour matches established mechanics	☐	☐	☐	☐	☒
instruction following	answers the question asked, not a nearby one	☐	☐	☐	☒	☐
harmlessness	following it would not risk an unsafe mix or test decision	☐	☐	☐	☐	☒
formatting	structure fits the length the answer needs	☐	☐	☒	☐	☐
verbosity	as short as it can be while staying complete	☐	☒	☐	☐	☐
tone	reads as a peer explaining, not a lecture	☐	☐	☐	☒	☐

the double arrow marks two dimensions in tension on the same response

Six axes, one judgement — and they conflict.

A response can be truthful and badly formatted, or fluent and wrong. Six axes, one judgement. **Learn these six** — each returns later in the course.

instrument (The instrument shown is synthetic, built for this course.)

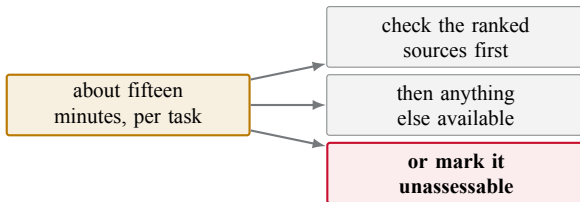
The person is made to attempt the task before judging it



The pipeline buys the attempt in order to get a reliable verdict. It is the most expensive step in the job and it is required in three of the five documents.

instrument (Three of five documents.)

Verification is time-boxed at about fifteen minutes, against a ranked source list



A claim that would take longer than the box to check does not get checked. It gets a band of its own.

instrument (One of five documents, stated as a rule.)

Two incompatible answers to who is qualified to judge

SCHEMATIC — both columns invented for this course.

generalist raters

who rates

a hired generalist rater

what 'good' means

matches a written style specification

constraint

prose fixed down to which punctuation is allowed

who sets the task

someone else supplies the question

domain professionals

who rates

a practising domain professional

what 'good' means

matches the professional's own working judgement

constraint

near-total discretion over the answer's form

who sets the task

the professional designs their own question

Same vocabulary describes both jobs. Both appear in the same set of review documents. They are not reconcilable.

Both are described with the same vocabulary. They are not the same job, and no single description of "the annotator" covers both.

instrument (Corroborated across documents.)

Substrate

Formation

Control

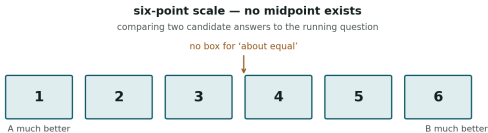
Measure

Failure

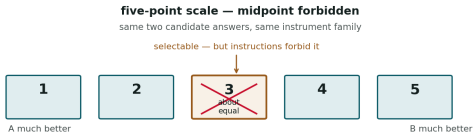
Limits

Equivalence is engineered out of the data, by opposite mechanisms

ILLUSTRATIVE — both scales invented for this course.



equivalence engineered out — structurally



equivalence engineered out — by instruction

The resulting dataset therefore contains no ties, whatever the rater actually thought.

Two instruments, two opposite designs, the same result: the dataset contains no ties, whatever the rater actually thought.

instrument (Two of five documents.)

Substrate

Formation

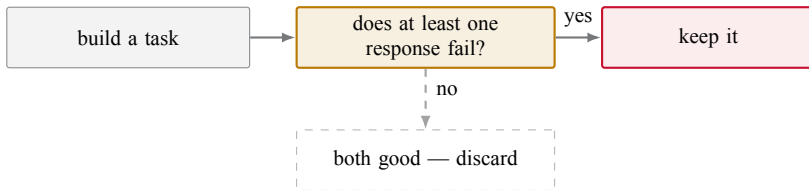
Control

Measure

Failure

Limits

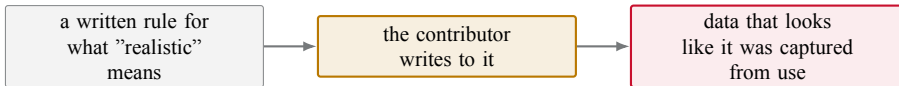
One instrument collects comparisons only where a model has already failed



The preference data from that instrument is conditioned on failure. It is not a sample of how the model behaves in use.

instrument (One of five documents.)

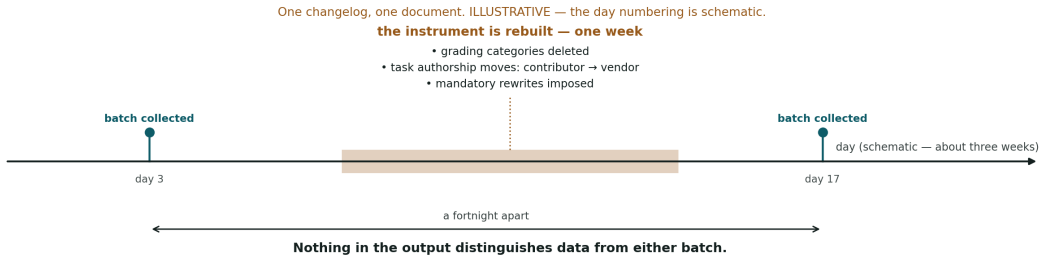
Realism in the data is manufactured to a specification



One line in one instrument, and it complicates the picture of preference data as behaviour observed in the wild.

instrument (One of five documents.)

One instrument was rebuilt in a single week, in the middle of collection

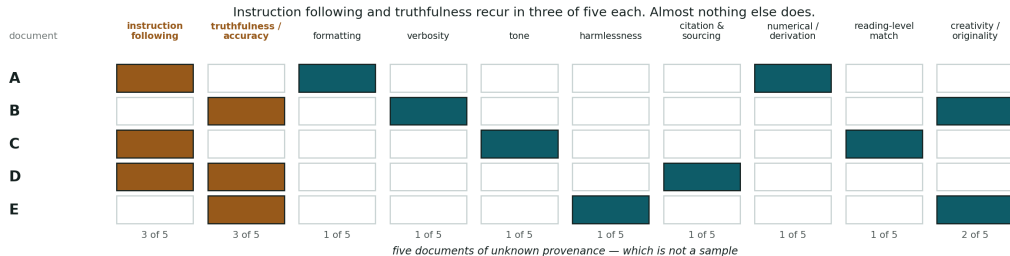


Grading categories deleted, task authorship moved, mandatory rewrites imposed. Data gathered a fortnight apart was graded by materially different devices, with nothing in the output to distinguish them.

instrument (One of five documents, from a dated changelog.)



The instruments diverge in structure far more than in vocabulary



An instruction-following dimension recurs, and a truthfulness dimension recurs. Almost nothing else does — not what else is measured, nor at what granularity, nor under what weighting.

instrument (Five documents of unknown provenance, which is not a sample.)

Substrate

Formation

Control

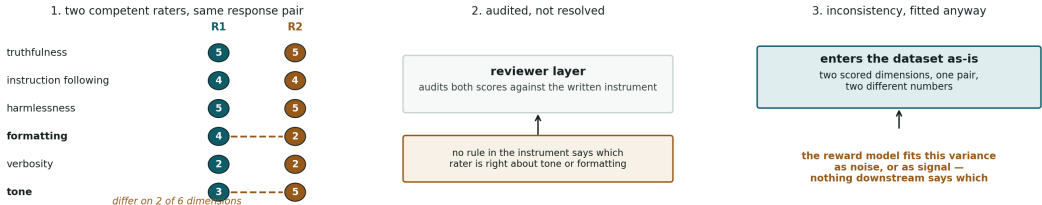
Measure

Failure

Limits

Two competent people disagree, and the disagreement goes into the data

ILLUSTRATIVE — scores and reviewer note invented for this course.

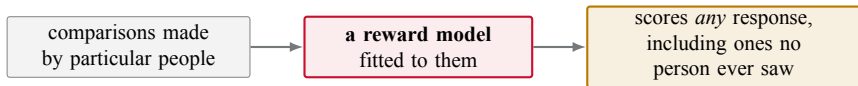


Where they diverge, the question is whether the rubric or the rater is at fault. An ambiguous dimension produces disagreement that no amount of rater training removes.

instrument (The audit layer is one of five documents; the downstream consequence is derived.)



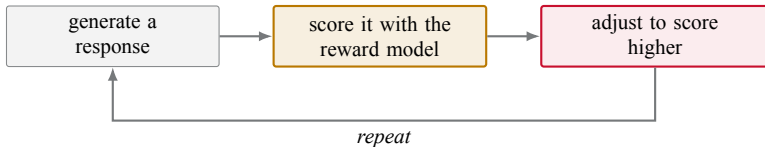
The reward model is a model of a rater



A finite set of comparisons, made by particular people, under particular instructions, becomes a function that returns a number for anything. **Hold on to that sentence.**

[V] Stiennon, N., Ouyang, L., Wu, J., et al., 2020. *preprint — venue unconfirmed*, arXiv:2009.01325.

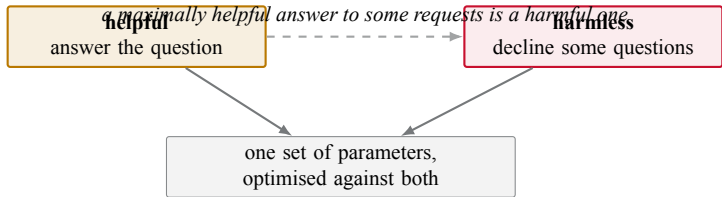
The model is then adjusted to score higher against that function



Generate, score, adjust, repeat. This loop is **reinforcement learning from human feedback**, shortened to **RLHF**. The reported result: outputs from a **1.3 billion** parameter fine-tuned model were preferred to those of a **175 billion** parameter base model. Credit the whole post-training pipeline — the comparison is fine-tuning *and* this step, not this step alone.

[V] Ouyang, L., Wu, J., Jiang, X., et al., 2022. *NeurIPS 2022*, arXiv:2203.02155.

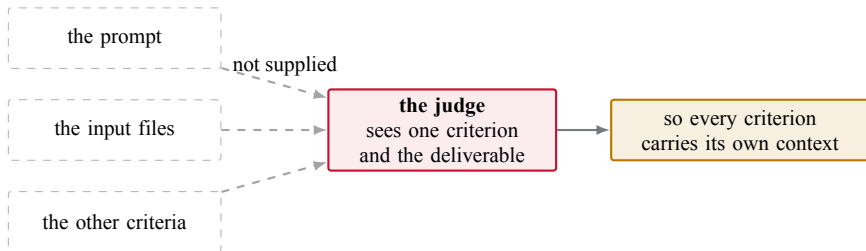
Helpful and harmless are collected separately, and they conflict



Refusal is trained in, from the same kind of human judgement as everything else. The tension is the method rather than a defect in it.

[V] Bai, Y., Jones, A., Ndousse, K., et al., 2022. *preprint*, arXiv:2204.05862.

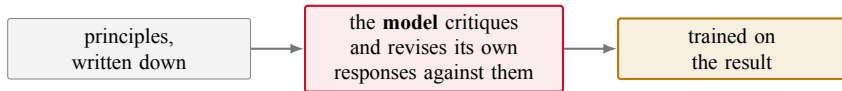
Where a model does the grading, every criterion must be checkable on its own



The blindness is deliberate, and it dictates how criteria are written: stripped of everything except the observable surface.

instrument (Two of five documents, and one states the causal direction outright.)

Part of the human layer is then replaced by the model itself



The named instance is **Constitutional AI**; the general pattern is **RLAIF** — reinforcement learning from *AI* feedback, with the model standing in for the rater. Human labour moves from judging comparisons to writing and maintaining the principles.

[V] Bai, Y., Kadavath, S., Kundu, S., et al., 2022. *preprint*, arXiv:2212.08073.

Behaviour traces back to the mechanism, and the traces are not equally strong

What you see	Where it comes from	Standing
Refuses some requests firmly	harmlessness trained as a competing objective	published
Agrees when you push back	preference data rewards agreement	plausible, not evidenced in these documents
Long, structured, confident answers	not known	the folk answer is contradicted

The verbosity row is the interesting one. The folk story says raters liked long answers, so models got long. One instrument breaks ties toward *brevity*; another *deleted* length as a grading category partway through. And models are verbose anyway.

derived (middle column argues; right column says how far it goes)



The pipeline aims deliberately at the band where the model believes it knows

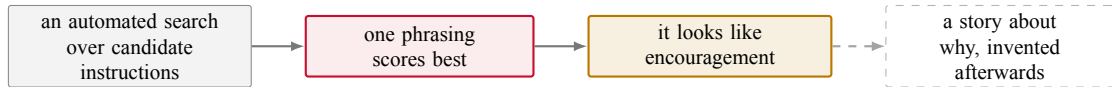


Contributors are instructed to build tasks in the middle band — not common knowledge, where the model is reliable; not genuinely obscure, where it declines. **That band is where your own literature sits.**

instrument (One of five documents, stated as an operational rule.)



Attribution goes wrong: a phrasing found by search, given a story afterwards

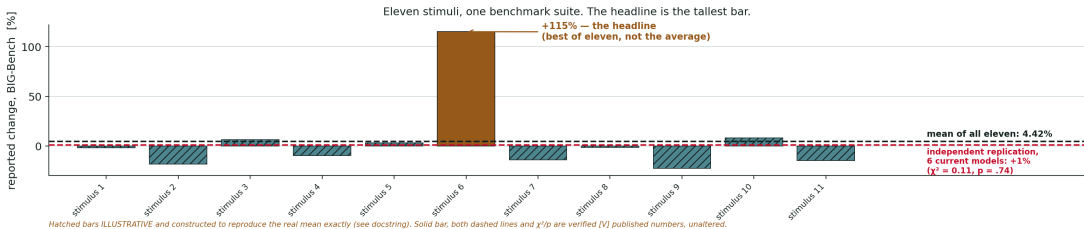


Reported gains up to 8% and up to 50% on two benchmarks, scored on one model. The winning instruction is specific to the model it was scored on; change the model and the best phrasing changes with it.

[V] Yang et al., 2023. *Google DeepMind; arXiv preprint (ICLR 2024 per the arXiv comment)*, arXiv:2309.03409.

[V] Ye, Q., Axmed, M., Pryzant, R. & Khani, F., 2024. *Findings of the ACL 2024*, arXiv:2311.05661.

Attribution goes wrong: the best of eleven, reported as the result

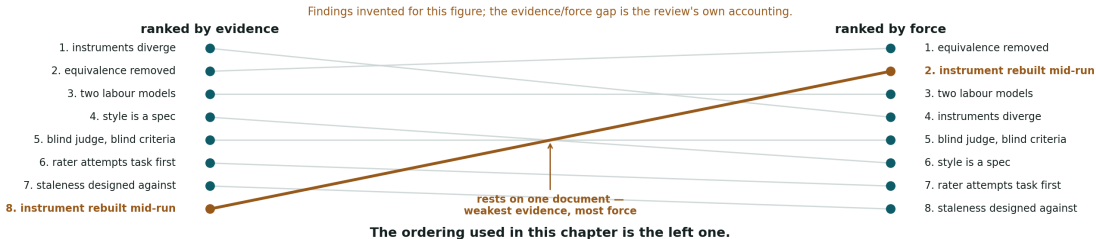


An independent replication across six current models found an insignificant increase of 1%. Re-deriving the original's own arithmetic gives **4.42%** averaged over all eleven stimuli, against the **115%** that became the headline.

[V] Li, C., Wang, J., Zhang, Y., et al., 2023. *technical report, not peer reviewed*, arXiv:2307.11760.

[V] Vaugrante, L., Niepert, M. & Hagendorff, T., 2024. *preprint — venue unconfirmed*, arXiv:2409.20303.

The order used here is by evidence, and it is deliberately not the order of interest



The finding this room is most likely to remember — an instrument rebuilt mid-collection — ranks eighth. It rests on one changelog. Both of those are true at once, and printing the count on every frame is what lets you hold them together.

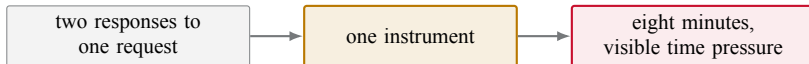
derived (The ranking is the instructor's, applied to fourteen findings.)

What I got wrong while assembling this chapter

The claim	What killed it
The rater's role migrates over time, from judge toward author	the earliest dated document already does both, and the only before-and-after evidence anywhere runs the other way
Three features are how instruments work	each appeared in exactly one document of five
The instruments share almost no vocabulary	too strong; two dimensions recur in three documents each
The person's repaired text becomes training data	two claims welded together; the labour recurs, the collection does not
The first survived two revisions before the evidence killed it, and it was the best story in the set.	

derived (A correction log kept while the findings were assembled.)

Now you do it



Your domain

mechanics, concrete or fluids,
so every one of you can check
the physics

Record confidence

a score per dimension, a
preference, and how sure you
are

Time-box the checking

over two minutes to verify, mark
it **unassessable**

instrument (The instrument is synthetic, built for this course on this room's domain.)

How the session runs

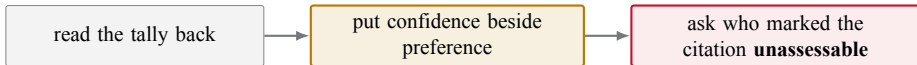
	Step	Minutes
1	Rate , individually, in silence, under visible time pressure	8
2	Collect the tally — preference, then confidence beside it	3
3	Reveal the physical error and the fabricated citation	5
4	Name what happened	5
5	Reviewer round — swap sheets, audit against the instrument	10
6	Adjudicate — the defensible grade per axis, including where the rubric is ambiguous	4

Steps 1 to 4 carry the lesson. With an odd number in the room, make one group of three at step 5 and rotate rather than swap.

no external source



What to say after the reveal



Whatever the tally says, **call it a tally**. Five or six raters is not a distribution, and this course does not present six points as one.

derived

What we covered, and what is still open

What we covered

- ▶ A predictor, plus demonstrations, plus preferences, plus optimisation against a model of those preferences
- ▶ The rating layer is smaller and more variable than *human preferences* suggests
- ▶ Behaviour traces to mechanism, and the traces are not equally strong
- ▶ An instrument that cannot record a tie produces a dataset without ties
- ▶ Where a machine grades, criteria are stripped to the observable surface

What is still open

- ▶ If instructions shaped it, which instructions steer it, and why those?
- ▶ How would you know a prompt change helped, rather than looked like it helped?
- ▶ How many raters would that tally have needed before it meant anything?
- ▶ Which behaviours on the mechanism table would survive better evidence?

derived